Carry Obstructions and Infinite Periodic-Crossing Families in a Decimal Digit-Sum Multiplier Problem

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Abstract

For a positive integer n, let $a(n)$ be the largest nonnegative integer k satisfying the decimal digit-sum equation $\mathrm{s}(kn) = k$, with $a(n) = 0$ if there is no positive solution. This is the multiplier formulation underlying OEIS A277223. We develop two structural mechanisms for this equation. *Carry obstruction* proves that the small Good multipliers $1, 2, 3, 4, 5, 6, 8, 10$ are necessarily nonterminal: a decimal rescaling, together with an exact carry-defect identity and first-carry localization, forces a larger Good multiplier. *Periodic crossing* studies the integer defect $\Delta_N(k) = \mathrm{s}(kN) - k$; if p is Good, digit-sum subadditivity makes Δ_N nonincreasing in every residue class modulo p.

The periodic method yields two explicit infinite families whose members are not obtained from one another by multiplication by a positive power of 10. For every $t \geq 0$ we construct $N_t^{(7)}$ with $a(N_t^{(7)}) = 7$ and a second family $N_t^{(11)}$ with $a(N_t^{(11)}) = 11$. The 7-family begins with the 152-digit witness

$$\frac{5 \cdot 10^{152} + 11}{7}.$$

For 11 we also record the 52-digit witness

$$\frac{9 \cdot 10^{52} + 2}{11},$$

while the infinite family is chosen so that all ten nonzero residue classes cross at the same pure decimal boundary. As an application we obtain the exact small-value spectrum

$$\{a(n) : a(n) < 12\} = \{0, 7, 9, 11\},$$

thereby refuting the OEIS conjecture that only 0 and 9 occur below 12. An accompanying Lean 4/Mathlib development checks the small-spectrum theorem and both parameterized families under a pinned toolchain. The mathematical argument below is self-contained; the formalization is a supplementary verification layer.

Keywords. decimal digit sum; carry propagation; periodic decimal blocks; infinite families; OEIS A277223; Lean 4.

1 Introduction

Let $\mathrm{s}(m)$ denote the sum of the decimal digits of a nonnegative integer m. We call a multiplier k *Good for n* when

$$\mathrm{Good}(n, k) \iff \mathrm{s}(kn) = k. \tag{1}$$

Since 0 is always Good and lemma 2.1 shows that only finitely many positive multipliers are Good for a fixed n, we define

$$a(n) = \max\{k \geq 0 : \mathrm{Good}(n, k)\}.$$

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The sequence A277223 is described in terms of this largest multiplier. The OEIS entry records the conjecture that if this largest value is below 12, then it is either 0 or 9 [1]. The purpose of this paper is not only to refute that statement but to determine the complete spectrum below 12.

The paper has two levels of conclusions. The structural contribution is the existence of infinite terminal families for the values 7 and 11 whose distinct members are not related by multiplication by a positive power of 10; the OEIS small-value problem is then a finite-spectrum application. Finite-state methods for positional numeration systems are classical [3, 4]; our regularity statement below is a specialized carry-transducer observation, while the main arguments use explicit carry obstructions and periodic defect crossings rather than generic automaton search.

Theorem 1.1 (Infinite periodic terminal families)**.** *There are explicit sequences* $(N_t^{(7)})_{t\geq 0}$ *and* $(N_t^{(11)})_{t\geq 0}$ *such that*

$$a(N_t^{(7)}) = 7, \qquad a(N_t^{(11)}) = 11 \qquad (t \geq 0).$$

Within each sequence no two members differ by multiplication by a positive power of 10.

Theorem 1.2 (Exact spectrum below twelve)**.** *For the decimal digit-sum multiplier function* $a(n)$*,*

$$\{a(n) : a(n) < 12\} = \{0, 7, 9, 11\}.$$

In particular, $1, 2, 3, 4, 5, 6, 8, 10$ *never occur as maximal Good multipliers.*

The proof is funneled through two reusable modules. Carry Obstruction shows that every Good multiplier in $\{1, 2, 3, 4, 5, 6, 8, 10\}$ is nonmaximal. Periodic Crossing proves uniqueness of a prescribed Good multiplier from one skipped zero crossing in each nonzero residue class. The parameterized constructions choose their block counts so that these crossings occur at controlled decimal scales, making the 152-digit witness the first member of a family and showing independently that the value 11 also occurs in an infinite periodic family. The 52-digit witness serves as a separate explicit example.

2 Preliminaries on decimal digit sums

Lemma 2.1 (Finiteness of Good multipliers)**.** *For every fixed* $n \geq 1$*, only finitely many positive* k *satisfy* $\mathrm{Good}(n, k)$*. Since* $\mathrm{Good}(n, 0)$ *always holds, the set of nonnegative Good multipliers is finite and nonempty; hence* $a(n)$ *is well-defined.*

Proof. Let $k > 0$ be Good and let L be the number of decimal digits of kn. Then

$$k = \mathrm{s}(kn) \leq 9L, \qquad 10^{L-1} \leq kn \leq 9Ln.$$

For fixed n, the last inequality can hold for only finitely many L: indeed $b_L = 10^{L-1}/L$ satisfies $b_{L+1}/b_L = 10L/(L+1) \geq 5$ for $L \geq 1$, so (b_L) is unbounded. Thus L, and consequently $k \leq 9L$, is bounded. □

2.1 Block separation and subadditivity

We use two elementary facts repeatedly.

Lemma 2.2 (Separated blocks)**.** *If* $0 \leq x < 10^w$*, then*

$$\mathrm{s}(x + 10^w y) = \mathrm{s}(x) + \mathrm{s}(y).$$

Proof. The summands occupy disjoint decimal columns. Equivalently, the decimal digit list of the sum is obtained by padding the digit list of x to width w and appending the digit list of y. □

Lemma 2.3 (Subadditivity)**.** *For all $x, y \geq 0$,*

$$s(x+y) \leq s(x) + s(y).$$

Proof. Add x and y column by column. Each unit of carry replaces ten units in one column by one unit in the next and therefore decreases the total digit sum by 9. Hence carries can only decrease the digit sum relative to the carry-free sum. □

2.2 The carry-defect identity

The preceding proof has a useful exact form. Let

$$m = \sum_{i=1}^{r} d_i 10^{e_i}, \qquad 1 \leq d_i \leq 9,$$

where the exponents e_i are distinct. Fix an integer $c \geq 1$. Write the decimal profile of cd as

$$cd = \sum_{u \geq 0} P_c(d,u) 10^u,$$

where all but finitely many $P_c(d,u)$ are zero. Before carries are propagated, column t of cm has load

$$L_t = \sum_i P_c(d_i, t - e_i). \tag{2}$$

Let C_t be the incoming carry at column t, with $C_0 = 0$, and let E_t be the final output digit. Thus

$$L_t + C_t = E_t + 10C_{t+1}, \qquad 0 \leq E_t \leq 9. \tag{3}$$

Only finitely many terms are nonzero if the carry tail is included.

Theorem 2.4 (Carry-defect identity)**.** *With the preceding notation,*

$$\sum_{i=1}^{r} s(cd_i) - s(cm) = 9 \sum_{t \geq 0} C_{t+1}. \tag{4}$$

Consequently, $s(cm) = \sum_i s(cd_i)$ *if and only if the superposition is carry-free.*

Proof. Summing (3) over all columns gives

$$\sum_t L_t + \sum_t C_t = \sum_t E_t + 10 \sum_t C_{t+1}.$$

Since $C_0 = 0$ and the terminal carry is included in the recurrence, the two carry sums differ only by the factor 9. Moreover,

$$\sum_t L_t = \sum_i \sum_u P_c(d_i, u) = \sum_i s(cd_i), \qquad \sum_t E_t = s(cm).$$

Substitution yields (4). □

Corollary 2.5 (First-carry locality)**.** *Suppose every cd_i has at most w decimal digits. If the superposition is not carry-free, then there is a first overloaded column whose contributing original digits lie within a window of at most w consecutive positions.*

Proof. At the first carry column t the incoming carry is zero, so (3) gives $L_t \geq 10$. A digit at exponent e_i can contribute to L_t only when $0 \leq t - e_i < w$. □

This locality is the reason the proof below is finite without enumerating integers n: only short collision motifs can obstruct a proposed rescaling.

3 Carry obstruction

3.1 Decimal rescaling

Assume that k is Good for n and put

$$m = kn, \qquad s(m) = k.$$

Definition 3.1 (Rescaling witness)**.** *A triple* (c, j, z) *is a rescaling witness for* (k, m) *if*

$$j > k, \qquad ck = j10^z, \qquad s(cm) = j. \tag{5}$$

Lemma 3.2 (Decimal rescaling)**.** *If* (c, j, z) *is a rescaling witness for* (k, m) *and* $m = kn$*, then* j *is Good for* n*.*

Proof. The scaling identity gives

$$cm = ckn = j\,10^z n.$$

Appending z decimal zeros does not change digit sum, hence

$$s(jn) = s(j\,10^z n) = s(cm) = j.$$

□

We call k *carry-obstructed* if every decimal integer m with $s(m) = k$ admits such a witness. A carry-obstructed k can never be the maximal Good multiplier.

3.2 A profile-safe criterion

Let $\lambda = (d_1, \ldots, d_r)$ be the multiset of nonzero digits of m. For fixed c, let w be at least the maximum length of the products cd_i. At a single output column the relative profile offsets contributed by distinct input positions are distinct. Define

$$M_c(\lambda) = \max\left\{\sum_{i \in I} P_c(d_i, u_i) : I \subseteq \{1, \ldots, r\},\ u_i \in \{0, \ldots, w-1\} \text{ distinct}\right\}. \tag{6}$$

Lemma 3.3 (Profile-safe criterion)**.** *If*

$$M_c(\lambda) \le 9 \quad \textit{and} \quad \sum_i s(cd_i) = j,$$

then every placement of the digits with multiset λ *satisfies*

$$s(cm) = j.$$

Proof. The definition of M_c bounds every raw column load L_t in (2) by 9. Hence no first carry can occur. The conclusion follows from theorem 2.4. □

3.3 The values 1 through 6

The first six exclusions are immediate manifestations of the same principle.

Proposition 3.4. *None of* $1, 2, 3, 4, 5, 6$ *is a maximal Good multiplier.*

Proof. Let $m = kn$ with $\mathrm{s}(m) = k$.

If $1 \le k \le 4$, every decimal digit of m is at most 4. Doubling is carry-free, so

$$\mathrm{s}(2m) = 2\,\mathrm{s}(m) = 2k.$$

Thus $(2, 2k, 0)$ is a rescaling witness.

For $k = 5$, if all digits are at most 4, the same argument gives the larger Good multiplier 10. Otherwise a digit 5 occurs, and since the total digit mass is 5 one has $m = 5 \cdot 10^e$. The witness $(18, 9, 1)$ gives $18m = 90 \cdot 10^e$.

For $k = 6$, the carry-free doubling case again gives 12. If a digit 6 occurs, then $m = 6 \cdot 10^e$ and $(15, 9, 1)$ is a witness. The only remaining digit partition is $5 + 1$, so

$$m = 5 \cdot 10^e + 10^f, \qquad e \neq f.$$

With $(c, j, z) = (25, 15, 1)$ the two product blocks are 125 and 25. Their digit sums are 8 and 7, and for distinct shifts they remain carry-free (the adjacent placements are 375 and 1275). Hence $\mathrm{s}(25m) = 15$. □

3.4 The value 8: a depth-two obstruction tree

Let $\mathrm{s}(m) = 8$. If all digits are at most 4, doubling yields a larger Good multiplier 16. Otherwise the nonzero digit partition is one of

$$8,\ 7+1,\ 6+2,\ 6+1+1,\ 5+3,\ 5+2+1,\ 5+1+1+1. \tag{7}$$

There are 22 unordered nonzero decimal digit partitions of total mass 8; exactly seven contain a digit at least 5, namely the partitions in (7). Thus the carry-free doubling case together with the following seven rows is exhaustive. The complete carry-obstruction tree is summarized in table 1. A motif such as 53 means that the displayed digits occur in adjacent decimal positions, in the usual high-to-low order.

Digit partition	Primary witness (c, j, z)	First obstruction	Resolution
8	$(1125, 9, 3)$	none	direct
$7 + 1$	$(15, 12, 1)$	none	profile-safe
$6 + 2$	$(15, 12, 1)$	none	profile-safe
$6 + 1 + 1$	$(25, 20, 1)$	16	$(125, 10, 2)$
$5 + 3$	$(25, 20, 1)$	53	$(2125, 17, 3)$
$5 + 2 + 1$	$(25, 20, 1)$	52 or 12	52: $(2125, 17, 3)$ except 152, then $(125, 10, 2)$; 12: $(2250, 18, 3)$ except 512, then $(125, 10, 2)$
$5 + 1 + 1 + 1$	$(45, 36, 1)$	115	$(375, 30, 2)$ except 1151, then $(1750, 14, 3)$, or 1115, then $(2750, 22, 3)$

Table 1: Carry-obstruction tree for $k = 8$. Every terminal target j is strictly larger than 8.

For the first three rows the exact profile maxima are respectively $9, 6, 9$, and the sums of the individual product digit sums are $9, 12, 12$; hence these rows are profile-safe (the one-digit 8 row is immediate).

We record the local argument once, since it is representative of all rows. For the partition $6 + 1 + 1$, multiplication by 25 produces the profiles

$$150, \qquad 25, \qquad 25,$$

whose carry-free digit-sum total is 20. The only possible first overload is $5+5$; by corollary 2.5 it occurs exactly when one of the 1 digits sits immediately above the 6, i.e. the motif 16. Then

$$m = 16 \cdot 10^a + 10^b, \qquad b \notin \{a, a+1\}.$$

Multiplication by 125 gives blocks 2000 and 125. Every allowed relative shift is carry-free, and the digit-sum total is $2+8=10$.

Completeness of the remaining local checks is recorded explicitly in Appendix B. The point is not an unbounded case split: by corollary 2.5, once the relative shift exceeds the combined profile width the blocks are separated, so only the listed overlapping shifts exist. For example, the primary profiles for $5+2+1$ are $125, 50, 25$, whose only first-overload motifs are 52 and 12. In the 52 branch, the secondary profiles 110500 and 2125 fail only at the admissible relative placement producing 152; in the 12 branch, 27000 and 11250 fail only at the placement producing 512. For $5+1+1+1$, after the unique primary motif 115, the secondary profiles 43125 and 375 have exactly two admissible exceptional shifts, producing 1151 and 1115. The terminal arithmetic is

$$\begin{aligned} 125 \cdot 152 &= 19000, & 125 \cdot 512 &= 64000, \\ 1750 \cdot 1151 &= 2014250, & 2750 \cdot 1115 &= 3066250, \end{aligned}$$

with digit sums $10, 10, 14, 22$, respectively. Hence every branch ends in a larger Good multiplier.

Proposition 3.5. *The value* 8 *is carry-obstructed and therefore is never maximal.*

Proof. If every digit is at most 4, the carry-free doubling witness gives target 16. Otherwise (7) is exhaustive: for a leading digit $8, 7, 6, 5$, the remaining masses are respectively $0, 1, 2, 3$, giving $1+1+2+3=7$ high-digit partitions. The first three rows of table 1 are profile-safe. In the remaining four rows, corollary 2.5 reduces every failure of the primary witness to the listed first-overload motifs. Appendix B checks all admissible overlapping placements of the secondary product blocks; outside those finite ranges the blocks are separated by lemma 2.2. Every exceptional placement is terminated by one of the displayed witnesses with target $j > 8$. Hence every m with $\mathrm{s}(m) = 8$ admits a rescaling witness to a larger Good multiplier. ◻

3.5 The value 10

If 10 is Good for n, then

$$\mathrm{s}(10n) = \mathrm{s}(n) = 10.$$

Thus the input itself has digit mass 10. If all digits are at most 4, doubling gives target 20. If at least two digits are at least 5, the only possibility is $5+5$. Multiplication by 12 gives two shifted 60 blocks; they are separated unless the two input positions are adjacent, in which case their sum is 660. In every case the digit sum is 12.

Assume therefore that exactly one digit is at least 5. If that digit is $9, 8, 7, 6, 5$, the remaining mass is respectively $1, 2, 3, 4, 5$ and is partitioned using parts at most 4. The corresponding counts are $1, 2, 3, 5, 6$, for a total of 17. Fifteen are position-independent under the profile-safe criterion and fall into the five witness families in table 2; the remaining two are $5+4+1$ and $5+3+2$, treated immediately afterward.

For every row the sum of individual product digit sums equals c and the profile maximum (6) is at most 9. For instance,

$$22 \cdot 6 = 132, \quad 22 \cdot 3 = 66, \quad 22 \cdot 1 = 22,$$

with digit-sum total $6+12+4=22$ and profile maximum 9.

Only two partitions are not covered by the table. For $5+4+1$, the primary witness $c = 21$ has profiles $105, 84, 21$ and first-carry motifs 54 and 145. The motif 54 is resolved by $c = 24$ because

$$24 \cdot 54 = 1296, \qquad 24 \cdot 1 = 24,$$

$c = j$	Profile-safe digit partitions of total mass 10	M_c
12	$9+1$	9
13	$8+2$; $8+1+1$	7; 5
15	$7+3$; $7+2+1$; $6+4$; $6+2+2$	9; 9; 9; 9
21	$7+1+1+1$; $6+1+1+1+1$; $5+2+2+1$; $5+2+1+1+1$; $5+1+1+1+1+1$	9; 8; 9; 9; 7
22	$6+3+1$; $6+2+1+1$; $5+3+1+1$	9; 7; 9

Table 2: Profile-safe families for $k = 10$. The final column gives the exact maximum raw column load $M_c(\lambda)$. Since $c \cdot 10 = c \cdot 10^1$, each c is itself the larger Good target.

whereas 145 is resolved by

$$12 \cdot 145 = 1740, \qquad \mathrm{s}(1740) = 12.$$

For $5+3+2$, the primary profiles 105, 63, 42 have the unique first motif 53. Multiplication by 24 then produces 1272 and 48; the sole secondary collision is 253, for which

$$12 \cdot 253 = 3036, \qquad \mathrm{s}(3036) = 12.$$

Thus the obstruction tree again has depth at most two.

Proposition 3.6. *The value* 10 *is carry-obstructed and therefore is never maximal.*

Proof. The three initial cases are exhaustive: either all digits are at most 4, at least two digits are at least 5 (forcing $5+5$), or exactly one digit is at least 5. In the last case the preceding count gives exactly 17 partitions. The fifteen rows of table 2 are profile-safe. The only remaining partitions are $5+4+1$ and $5+3+2$; first-carry locality gives exactly the motifs stated above, and Appendix B checks every admissible overlapping placement of the secondary blocks. Their sole unresolved secondary collision is 253, which has the terminal target 12. Thus every m with $\mathrm{s}(m) = 10$ has a rescaling witness to some $j > 10$. □

Combining propositions 3.4 to 3.6 gives the first general module.

Theorem 3.7 (Carry Obstruction theorem)**.** *If*

$$k \in \{1, 2, 3, 4, 5, 6, 8, 10\}$$

and k is Good for n, then there exists a Good multiplier $j > k$. Consequently none of these eight values can equal $a(n)$.

Proof. Apply propositions 3.4 to 3.6 to $m = kn$, for which $\mathrm{s}(m) = k$, and then use the decimal-rescaling lemma lemma 3.2. □

3.6 Finite-state scope and scalability

The local carry argument is not confined to $k < 12$. For a fixed multiplier c, right-to-left schoolbook multiplication of an arbitrary decimal input requires only the current carry and the accumulated output digit sum. If one is testing whether the final digit sum equals a fixed target j, the accumulated sum may be capped at $j + 1$. Thus one lane has at most

$$c(j + 2)$$

states: the carry is $< c$ and the capped score lies in $\{0, \ldots, j + 1\}$. A finite collection of Good/Not-Good conditions is therefore recognized by a finite product automaton.

Proposition 3.8 (Regularity of fixed multiplier constraints)**.** *For every fixed c, j, the set of finite decimal digit strings representing integers n with $\mathrm{s}(cn) = j$ is a regular language. More generally, every finite Boolean combination of conditions $\mathrm{s}(c_i n) = j_i$ is decidable by a finite-state transducer.*

Proof. Scan the digits of n from least to most significant. For one condition keep the state (γ, s), where $0 \le \gamma < c$ is the multiplication carry and $s \in \{0, \dots, j+1\}$ is the output digit sum capped at $j+1$. On input digit d, write $cd + \gamma = e + 10\gamma'$ with $0 \le e \le 9$ and update to $(\gamma', \min(j+1, s+e))$. After the last input digit, add the decimal digit sum of the remaining carry and accept exactly when the total is j. This gives a finite deterministic automaton. Products and complements handle finite Boolean combinations. □

Remark 3.9. *The proposition proves decidability, not polynomial scalability in the number of simultaneous multipliers: a naive product construction grows multiplicatively. The obstruction trees used for* 8 *and* 10 *exploit digit mass and first-carry locality to avoid that generic state explosion. Thus the finite-state model is a completeness backstop, whereas the paper's main proofs remain structural.*

4 Periodic crossing

4.1 Defect monotonicity

For fixed N, define the integer-valued defect

$$\Delta_N(k) = \mathrm{s}(kN) - k. \tag{8}$$

Thus k is Good exactly when $\Delta_N(k) = 0$.

Theorem 4.1 (Residue-class monotonicity)**.** *If p is Good for N, then*

$$\Delta_N(k+p) \le \Delta_N(k)$$

for every $k \ge 0$. Consequently, for each residue r, the sequence

$$q \longmapsto \Delta_N(pq + r)$$

is nonincreasing.

Proof. By lemma 2.3 and $\mathrm{s}(pN) = p$,

$$\begin{aligned}\Delta_N(k+p) &= \mathrm{s}(kN + pN) - k - p \\ &\le \mathrm{s}(kN) + \mathrm{s}(pN) - k - p \\ &= \Delta_N(k).\end{aligned}$$

Iteration gives the residue-class statement. □

Theorem 4.2 (Crossing criterion)**.** *Let $p > 0$ be Good for N. Suppose*

$$\Delta_N(2p) < 0,$$

and for every $1 \le r < p$ there is an integer $q_r \ge 0$ such that

$$\Delta_N(pq_r + r) > 0, \qquad \Delta_N(p(q_r+1) + r) < 0.$$

Then p is the unique positive Good multiplier for N.

Proof. In the zero residue class, monotonicity and $\Delta_N(2p) < 0$ exclude every pq with $q \ge 2$; p itself is Good. In a nonzero residue class, all indices up to q_r have defect at least the positive left endpoint, while all indices from $q_r + 1$ onward have defect at most the negative right endpoint. Hence no defect can vanish. □

4.2 Periodic blocks

For a width d, a low block W, a block count M, and a high block H, define recursively

$$\mathcal{B}^{(0)}_{W,d}(H) = H, \qquad \mathcal{B}^{(M+1)}_{W,d}(H) = W + 10^d \mathcal{B}^{(M)}_{W,d}(H).$$

Let

$$\mathrm{Rep}_{W,d}(M) = \mathcal{B}^{(M)}_{W,d}(0).$$

If $W < 10^d$, repeated application of lemma 2.2 gives

$$\mathrm{s}\Big(\mathcal{B}^{(M)}_{W,d}(H)\Big) = M\,\mathrm{s}(W) + \mathrm{s}(H). \tag{9}$$

We shall repeatedly use a local update of one aligned block. If $0 \le i < M$ and

$$W < 10^d, \qquad W + x < 10^d,$$

then

$$\mathrm{s}\Big(\mathrm{Rep}_{W,d}(M) + x10^{di}\Big) = (M-1)\,\mathrm{s}(W) + \mathrm{s}(W+x). \tag{10}$$

Indeed, split the repeated word immediately below and immediately above the i-th block. The three pieces are separated powers of 10^d; the middle block is changed from W to $W+x$, and the hypothesis $W + x < 10^d$ rules out an inter-block carry. More generally, if a tail $T < 10^s$ is placed below the repeated word, then

$$\mathrm{s}\Big(T + 10^s\big(\mathrm{Rep}_{W,d}(M) + x10^{di}\big)\Big) = \mathrm{s}(T) + (M-1)\,\mathrm{s}(W) + \mathrm{s}(W+x). \tag{11}$$

Thus an arbitrarily long periodic word can be modified at one aligned position using only the digit sums of W, $W+x$, and the fixed tail. This is the common local mechanism behind both infinite families below.

The following algebraic lemma is the source of the long periodic decimals.

Theorem 4.3 (Periodic-block normal form)**.** *Assume*

$$pQ + 1 = 10^d, \qquad pC = A10^t + B, \tag{12}$$

and set

$$N = C + 10^t\,\mathrm{Rep}_{AQ,d}(M).$$

Then

$$pN = A10^{dM+t} + B. \tag{13}$$

For a residue r, write

$$rA = ph_r + a_r, \qquad rC = h_r 10^t + T_r. \tag{14}$$

Then for every $q \ge 0$,

$$(pq+r)N = (Aq + h_r)10^{dM+t} + 10^t\,\mathrm{Rep}_{a_rQ,d}(M) + (Bq + T_r). \tag{15}$$

If $M = S + 1$, this can be regrouped as

$$(pq+r)N = L_{q,r} + 10^{d+t}\mathcal{B}^{(S)}_{W_r,d}(H_{q,r}), \tag{16}$$

where

$$W_r = a_r Q, \qquad H_{q,r} = Aq + h_r, \qquad L_{q,r} = W_r 10^t + Bq + T_r.$$

Whenever $W_r < 10^d$ and $L_{q,r} < 10^{d+t}$,

$$\mathrm{s}((pq+r)N) = \mathrm{s}(L_{q,r}) + S\,\mathrm{s}(W_r) + \mathrm{s}(H_{q,r}). \tag{17}$$

Proof. The identity $pQ+1=10^d$ yields, by induction on M,

$$p\,\mathrm{Rep}_{AQ,d}(M)+A=A10^{dM}. \tag{18}$$

Substitution into the definition of N gives (13). A second induction using $rA=ph_r+a_r$ gives

$$r\,\mathrm{Rep}_{AQ,d}(M)+h_r=h_r10^{dM}+\mathrm{Rep}_{a_rQ,d}(M). \tag{19}$$

Expanding $(pq+r)N=q(pN)+rN$ and using (14) and (19) yields (15). Splitting the lowest copy of W_r gives (16); (17) follows from separated blocks and (9). □

5 Two infinite periodic-crossing families

The periodic construction is not limited to isolated witnesses for the values 7 and 11. We choose the block counts so that the residue-class crossing occurs at a prescribed decimal scale. This removes the structural dependence on isolated long constants and gives infinite families in which no member is obtained from another merely by appending decimal zeros. The exponents 152 and 52 give explicit witnesses, with the former equal to the first member of the 7-family.

5.1 An infinite family with maximal multiplier 7

For $t\geq 0$, put

$$u_t=6t+2,\qquad M_t^{(7)}=\frac{7\cdot 10^{u_t}-25}{27}. \tag{20}$$

The quotient is integral: $10^6\equiv 1 \pmod{27}$ and the case $t=0$ is $7\cdot 10^2-25=675$. Equivalently,

$$M_0^{(7)}=25,\qquad M_{t+1}^{(7)}=10^6M_t^{(7)}+925925. \tag{21}$$

The defining balance relation is

$$7\cdot 10^{u_t}=27M_t^{(7)}+25. \tag{22}$$

Define

$$N_t^{(7)}=73+10^2\,\mathrm{Rep}_{714285,6}(M_t^{(7)})=\frac{5\cdot 10^{6M_t^{(7)}+2}+11}{7}. \tag{23}$$

Thus

$$7N_t^{(7)}=5\cdot 10^{6M_t^{(7)}+2}+11,$$

so 7 is Good for every family member.

For $1\leq r\leq 6$, write

$$5r=7h_r+a_r,\qquad 73r=100h_r+T_r,\qquad W_r=142857a_r.$$

The residue data are

r	h_r	a_r	T_r	W_r	$s(W_r)$
1	0	5	73	714285	27
2	1	3	46	428571	27
3	2	1	19	142857	27
4	2	6	92	857142	27
5	3	4	65	571428	27
6	4	2	38	285714	27

Let $R_{r,t}$ denote the lower periodic part of $rN_t^{(7)}$, so that

$$rN_t^{(7)} = h_r 10^{6M_t^{(7)}+2} + R_{r,t}.$$

Every six-digit block of $R_{r,t}$ is one of the six cyclic permutations displayed above and therefore uses only the digits $1, 2, 4, 5, 7, 8$. Also

$$\mathrm{s}(T_r - 11) = \mathrm{s}(T_r) - 2.$$

Since $M_t^{(7)} > t$, the correction

$$11 \cdot 10^{u_t} = 10^2 \big(11 \cdot 10^{6t}\big)$$

is aligned with the repeated width-six block of zero-based index t. Since $M_t^{(7)} > t$, that block lies within the periodic region. Adding the correction therefore replaces one block W_r by $W_r + 11$; the residue table gives $W_r + 11 < 10^6$ and $\mathrm{s}(W_r + 11) = 29$, so the update creates no inter-block carry. Consequently

$$\mathrm{s}(R_{r,t} + 11 \cdot 10^{u_t}) = 27M_t^{(7)} + \mathrm{s}(T_r) + 2, \tag{24}$$

$$\mathrm{s}(R_{r,t} + 11(10^{u_t} - 1)) = 27M_t^{(7)} + \mathrm{s}(T_r). \tag{25}$$

The second identity follows by first replacing T_r by $T_r - 11$ and then adding the separated block $11 \cdot 10^{u_t}$.

The finite residue table also satisfies

$$h_r + \mathrm{s}(T_r) - r = 9 \qquad (1 \le r \le 6). \tag{26}$$

Putting $q_t = 10^{u_t}$ into the exact residue normal form gives

$$\mathrm{s}((7q + r)N_t^{(7)}) = \mathrm{s}(5q + h_r) + \mathrm{s}(R_{r,t} + 11q).$$

Now

$$\mathrm{s}(5q_t + h_r) = 5 + h_r, \qquad \mathrm{s}(5(q_t - 1) + h_r) = 9u_t + h_r.$$

Using (22), (24)–(26), one obtains the uniform endpoint identities

$$\Delta_{N_t^{(7)}}(7(q_t - 1) + r) = 9u_t - 9 = 54t + 9 > 0, \qquad \Delta_{N_t^{(7)}}(7q_t + r) = -9 < 0 \tag{27}$$

for every $1 \le r \le 6$. The zero residue class satisfies

$$\Delta_{N_t^{(7)}}(14) = -9.$$

The crossing criterion therefore gives the first infinite-family theorem. Distinct members of the structural families below are additionally required not to differ by multiplication by a positive power of 10; this separates the constructions from the decimal scaling in Corollary 7.1.

Theorem 5.1 (Infinite terminal family for 7)**.** *For every $t \ge 0$,*

$$a(N_t^{(7)}) = 7.$$

In particular, there are infinitely many integers n with $a(n) = 7$, no two of which differ by multiplication by a positive power of 10.

Proof. Equation (27) gives a strict adjacent crossing in every nonzero residue class modulo 7, while $\Delta(14) < 0$ eliminates all later points in the zero residue class. Apply theorem 4.2. The recurrence (21) makes the block counts strictly increasing, and every $N_t^{(7)}$ ends in 73; hence distinct family members cannot differ by multiplication by a positive power of 10. □

The 152-digit counterexample is the first member: $M_0^{(7)} = 25$ and

$$N_0^{(7)} = \frac{5 \cdot 10^{152} + 11}{7}.$$

5.2 An infinite family with maximal multiplier 11

For $t \geq 0$, set

$$q_t = 100^{t+2} \tag{28}$$

and

$$M_t^{(11)} = \frac{11q_t - 29}{9}. \tag{29}$$

This is an integer because $q_t \equiv 1 \pmod 9$. Equivalently,

$$M_0^{(11)} = 12219, \qquad M_{t+1}^{(11)} = 100M_t^{(11)} + 319. \tag{30}$$

In particular $M_t^{(11)} \geq t+2$. The balance relation is simply

$$9M_t^{(11)} = 11q_t - 29. \tag{31}$$

Define

$$N_t^{(11)} = 82 + 10^2\,\mathrm{Rep}_{81,2}(M_t^{(11)}) = \frac{9 \cdot 10^{2M_t^{(11)}+2} + 2}{11}. \tag{32}$$

The periodic-block identity gives

$$11N_t^{(11)} = 9 \cdot 10^{2M_t^{(11)}+2} + 2, \tag{33}$$

so 11 is Good for every t.

For $1 \leq r \leq 10$, use the same residue decomposition

$$9r = 11h_r + a_r, \qquad 82r = 100h_r + T_r, \qquad W_r = 9a_r. \tag{34}$$

The complete residue data, together with the only local digit-sum changes needed at the crossing, are

r	h_r	T_r	W_r	$s(T_r)$	$s(T_r-2)$	$s(W_r+2)-9$	class
1	0	82	81	10	8	2	A
2	1	64	63	10	8	2	A
3	2	46	45	10	8	2	A
4	3	28	27	10	8	2	A
5	4	10	09	1	8	−7	B
6	4	92	90	11	9	2	A
7	5	74	72	11	9	2	A
8	6	56	54	11	9	2	A
9	7	38	36	11	9	2	A
10	8	20	18	2	9	−7	B

Thus class B consists exactly of residues 5 and 10. This two-class split is forced by the modulus-11/base-100 transfer and is not created by tuning q_t. Indeed, $a_r \equiv 9r \pmod{11}$ and $1 \leq a_r \leq 10$, so as r runs through $1, \ldots, 10$, the values a_r run through all nonzero residues modulo 11. Consequently $W_r = 9a_r$ runs through the ten blocks $09, 18, \ldots, 90$ in a permuted order. The aligned update $W_r \mapsto W_r + 2$ changes the digit sum by +2 for eight of these blocks, but necessarily crosses a decimal carry threshold at exactly 09 and 18, where the change is −7. These are precisely $r = 5$ and $r = 10$. Thus the A/B dichotomy is intrinsic to this local $p = 11$ transfer; the choice $q_t = 100^{t+2}$ only aligns the same +2 update with a repeated block. By contrast, the width-six $p = 7$ transfer above has a uniform local digit-sum increment in all six nonzero residue classes.

Let

$$R_{r,t} = T_r + 100\,\mathrm{Rep}_{W_r,2}(M_t^{(11)}),$$

so that the residue normal form is

$$(11q + r)N_t^{(11)} = (9q + h_r)10^{2M_t^{(11)}+2} + (R_{r,t} + 2q). \tag{35}$$

At the chosen crossing $q = q_t = 100^{t+2}$, the factor 100 in the definition of $R_{r,t}$ means that $2q_t$ updates the repeated block with zero-based index $t + 1$ (equivalently, the $(t + 2)$-nd repeated block counted from the low end). Since $M_t^{(11)} \geq t + 2$ and every $W_r + 2 < 100$, this block exists, changes from W_r to $W_r + 2$, and produces no carry. Hence

$$\mathrm{s}(R_{r,t} + 2q_t) = 9M_t^{(11)} + \mathrm{s}(T_r) + \mathrm{s}(W_r + 2) - 9. \tag{36}$$

Likewise

$$2(q_t - 1) = 2q_t - 2.$$

Every $T_r \geq 10$, so the subtraction of 2 is confined to the low tail, again without borrow into the periodic word. Therefore

$$\mathrm{s}(R_{r,t} + 2(q_t - 1)) = 9M_t^{(11)} + \mathrm{s}(T_r - 2) + \mathrm{s}(W_r + 2) - 9. \tag{37}$$

No unbounded carry analysis is hidden here: both identities are direct consequences of separated two-digit blocks.

The high blocks are equally simple. Since $q_t = 10^{2t+4}$ and $0 \leq h_r \leq 8$,

$$\mathrm{s}(9q_t + h_r) = 9 + h_r, \qquad \mathrm{s}(9(q_t - 1) + h_r) = 18(t + 2) + h_r. \tag{38}$$

For class A the residue table gives

$$h_r + \mathrm{s}(T_r) + \big(\mathrm{s}(W_r + 2) - 9\big) - r = 11,$$

$$h_r + \mathrm{s}(T_r - 2) + \big(\mathrm{s}(W_r + 2) - 9\big) - r = 9,$$

whereas for class B the corresponding constants are -7 and 0. Combining these identities with (31) yields, for every $t \geq 0$,

$$\Delta_{N_t^{(11)}}(11(q_t - 1) + r) = 18t + 27 > 0, \qquad \Delta_{N_t^{(11)}}(11q_t + r) = -9 < 0 \quad (r \notin \{5, 10\}), \tag{39}$$

and

$$\Delta_{N_t^{(11)}}(11(q_t - 1) + r) = 18t + 18 > 0, \qquad \Delta_{N_t^{(11)}}(11q_t + r) = -27 < 0 \quad (r \in \{5, 10\}). \tag{40}$$

Thus all ten nonzero residue classes cross strictly between the *same adjacent quotients* $q_t - 1$ and q_t.

The zero residue is immediate from (33):

$$22N_t^{(11)} = 18 \cdot 10^{2M_t^{(11)}+2} + 4, \qquad \Delta_{N_t^{(11)}}(22) = 13 - 22 = -9.$$

We have therefore proved the second family theorem.

Theorem 5.2 (Infinite terminal family for 11)**.** *For every $t \geq 0$,*

$$a(N_t^{(11)}) = 11.$$

Consequently there are infinitely many integers n with $a(n) = 11$, no two of which differ by multiplication by a positive power of 10.

Proof. Equations (39) and (40) give the required strict adjacent crossing in every nonzero residue class modulo 11, and $\Delta(22) = -9$ handles the zero residue class. Apply theorem 4.2. The recurrence (30) makes the block counts strictly increasing, and every $N_t^{(11)}$ ends in 82; hence no two members differ by multiplication by a positive power of 10. □

An explicit 52-digit witness is

$$N_{11}^{\mathrm{small}} = \frac{9 \cdot 10^{52} + 2}{11} = \underbrace{81\,81 \;\cdots\; 81}_{25 \text{ blocks}}\,82.$$

For completeness we also prove its maximality inside the manuscript.

Proposition 5.3 (Explicit 52-digit witness)**.** *For N_{11}^{small} above,*

$$a(N_{11}^{\mathrm{small}}) = 11.$$

Proof. The sparse identity

$$11N_{11}^{\mathrm{small}} = 9 \cdot 10^{52} + 2$$

shows that 11 is Good, while

$$\Delta_{N_{11}^{\mathrm{small}}}(22) = -9.$$

For $1 \le r \le 10$, let $q_r = 22$ for $r = 1, 2$ and $q_r = 21$ otherwise. Applying the periodic normal form with $M = 25$ leaves 24 complete two-digit residue blocks, each of digit sum 9, and gives the following endpoint defects:

r	1	2	3	4	5	6	7	8	9	10
q_r	22	22	21	21	21	21	21	21	21	21
$\Delta(11q_r + r)$	9	9	18	9	9	9	9	18	9	9
$\Delta(11(q_r + 1) + r)$	−9	−18	−9	−9	−9	−9	−9	−18	−9	−9

Every entry is obtained from at most the four low decimal digits together with the fixed contribution $24 \cdot 9 = 216$ of the untouched periodic blocks; thus the table is a direct finite arithmetic consequence of theorem 4.3, not an integer-range search. Each nonzero residue class has a strict adjacent sign crossing, and the zero residue class is negative from 22 onward. The crossing criterion theorem 4.2 therefore yields the claim. □

This witness gives a short explicit representative of the value 11; the infinite family above supplies the uniform pure-power crossing formula.

Remark 5.4 (Defects are multiples of nine)**.** *For either family, $p \in \{7, 11\}$ is coprime to 9 and $\mathrm{s}(pN) = p$, hence $N \equiv 1 \pmod 9$. Therefore*

$$\Delta_N(k) = \mathrm{s}(kN) - k \equiv kN - k \equiv 0 \pmod 9.$$

The multiples of nine in the crossing formulas are thus forced by the decimal congruence, not numerical coincidence.

6 Realizing 0 and 9

For completeness we prove the two classical surviving values internally.

Proposition 6.1. $a(1) = 9$.

Proof. The equation is $\mathrm{s}(k) = k$. Every one-digit $k \le 9$ satisfies it. If $k = 10q + r \ge 10$, then $q > 0$. Reading the decimal expansion gives $\mathrm{s}(q) \le q$, and hence

$$\mathrm{s}(k) = \mathrm{s}(q) + r \le q + r < 10q + r = k.$$

Thus 9 is maximal. □

Proposition 6.2. $a(62) = 0$.

Proof. Suppose $k > 0$ and $s(62k) = k$. Since $10 \equiv 1 \pmod 9$, every decimal integer is congruent modulo 9 to its digit sum. Thus

$$62k \equiv s(62k) = k \pmod 9,$$

so $61k \equiv 0 \pmod 9$. As $\gcd(61, 9) = 1$, one has $9 \mid k$.

Let L be the number of decimal digits of $62k$. Since a length-L decimal number has digit sum at most $9L$,

$$k \le 9L.$$

If $L \ge 5$, then

$$62k \le 558L < 10^{L-1},$$

where the last inequality follows from the base case $558 \cdot 5 < 10^4$ and the induction

$$558(L + 1) < 10 \cdot 558L.$$

This contradicts the fact that an L-digit positive integer is at least 10^{L-1}. Therefore $L \le 4$ and $k \le 36$. Together with $9 \mid k$, only

$$k \in \{9, 18, 27, 36\}$$

remain. Directly,

k	9	18	27	36
$62k$	558	1116	1674	2232
$s(62k)$	18	9	18	9

so none is Good. □

7 Proof of the classification

Proof of theorem 1.2. Let $a(n) < 12$. The only numerical possibilities are $0, 1, \ldots, 11$. The Carry Obstruction theorem theorem 3.7 removes

$$1, 2, 3, 4, 5, 6, 8, 10,$$

leaving only $0, 7, 9, 11$. Conversely, propositions 6.1 and 6.2 and theorems 5.1 and 5.2 realize all four values; for 7 and 11 one may take the first member of the corresponding infinite family. □

Corollary 7.1 (Infinite scaled families)**.** *For every $t \ge 0$,*

$$a(62 \cdot 10^t) = 0, \qquad a(N_0^{(7)} \cdot 10^t) = 7, \qquad a(10^t) = 9, \qquad a(N_0^{(11)} \cdot 10^t) = 11.$$

In particular, each of the four values in theorem 1.2 is attained infinitely often.

Proof. For every x and t, appending t trailing zeros gives

$$s(10^t x) = s(x).$$

Hence $\mathrm{Good}(10^t n, k)$ is equivalent to $\mathrm{Good}(n, k)$ for every k, and the entire set of Good multipliers is unchanged by the scaling $n \mapsto 10^t n$. Apply this to the four witnesses above. For each fixed positive witness, the integers obtained by multiplying by distinct powers of 10 are distinct. □

8 Supplementary Lean 4 formalization

An accompanying Lean 4/Mathlib development formalizes the definitions of Good and maximal Good multipliers, the small-value exclusion, defect monotonicity and the crossing criterion, the periodic-block machinery, the explicit witnesses, both parameterized infinite families, and the final spectrum theorem. For $k = 8$ and $k = 10$, the formal proof deliberately uses a second route: exact schoolbook-multiplication state certificates are generated as finite data and then checked by Lean with kernel reduction and a proved certificate-soundness theorem. Thus the Python generator is not a proof oracle. The repository pins Lean and Mathlib to version 4.32.2; its CI regenerates all certificate sources without drift, runs independent arithmetic checks, builds the complete project, and runs `leanchecker`. The formalization is supplementary: the mathematical proof in this article is self-contained, and the appendix provides only a navigation map between the two developments.

9 Discussion

Two points appear to be more general than this particular OEIS problem.

First, the Carry Obstruction method separates the *mass* constraint $\mathrm{s}(m) = k$ from the *geometry* of carries. For a fixed small mass, only finitely many nonzero digit partitions exist, while the first-carry theorem turns arbitrary digit spacing into bounded local motifs. This gives a systematic route for proving that a putative small solution cannot be maximal.

Second, Periodic Crossing replaces a global search for larger multipliers by a one-dimensional monotonicity argument in each residue class. Once a construction makes p Good, the inequality

$$\Delta_N(k + p) \le \Delta_N(k)$$

means that uniqueness is reduced to locating a single skipped zero crossing in each residue. Repetends arising from identities $pQ+1 = 10^d$ are particularly well suited to this purpose because multiplication has an exact block normal form.

The two infinite families show that the terminal values 7 and 11 are not isolated exceptions. A natural continuation is to characterize which target values admit an infinite periodic-crossing family and how the answer depends on the numeral base. The exact spectrum below 12 is therefore best viewed as the first finite application of the two structural mechanisms, rather than the endpoint of the method.

Data and code availability

The companion repository contains the Lean sources, certificate generator, independent arithmetic checks, CI configuration, and manuscript source. Reproducible machine verification is tied to the exact repository commit and its successful pinned CI run; the mathematical proofs in this article are self-contained.

About this project

This work determines the exact spectrum of maximal decimal digit-sum multipliers below 12 for the sequence underlying OEIS A277223 and proves that the terminal values 7 and 11 occur in explicit infinite periodic-crossing families. Its contributions are: a self-contained mathematical proof organized around two independent mechanisms (carry obstruction and periodic crossing); two concrete infinite families with arithmetic that can be checked by hand; and a supplementary Lean 4 formalization in which every layer—including the finite carry certificates—is re-verified by the Lean kernel and by an independent type checker, with a fully reproducible CI pipeline.

The proofs, the Lean formalization, and the verification infrastructure in the companion repository were developed with the assistance of GLM 5.3, a large language model trained by Z.ai. All mathematical claims were verified by the pinned Lean 4.32.2 kernel and re-checked by `leanchecker`; the AI system was used as a proof assistant, not as a trusted oracle.

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A Concrete arithmetic used by the carry trees

For reproducibility, the rescaling equalities used above are

$$
\begin{array}{r|c}
k = 8 & 1125 \cdot 8 = 9 \cdot 10^3, \quad 15 \cdot 8 = 12 \cdot 10, \quad 25 \cdot 8 = 20 \cdot 10, \\
 & 125 \cdot 8 = 10 \cdot 10^2, \quad 2125 \cdot 8 = 17 \cdot 10^3, \quad 2250 \cdot 8 = 18 \cdot 10^3, \\
 & 45 \cdot 8 = 36 \cdot 10, \quad 375 \cdot 8 = 30 \cdot 10^2, \quad 1750 \cdot 8 = 14 \cdot 10^3, \quad 2750 \cdot 8 = 22 \cdot 10^3; \\
k = 10 & 2 \cdot 10 = 20, \quad c \cdot 10 = c \cdot 10^1 \quad (c \in \{12, 13, 15, 21, 22, 24\}).
\end{array}
$$

The companion script `scripts/arithmetic_audit.py` independently recomputes these numerical identities and profile maxima as a regression check.

B Finite local completeness for the carry trees

This appendix records the finite checks that make the $k = 8$ and $k = 10$ obstruction trees exhaustive. By corollary 2.5, a failure of a proposed carry-free profile must have a first overloaded column, and only input digits lying inside the profile width can contribute to that column. Direct inspection of those bounded profiles gives exactly the following primary motifs:

Digit partition	primary product profiles	complete first-overload motifs
$8 : 6 + 1 + 1$	$150, 25, 25$	16
$8 : 5 + 3$	$125, 75$	53
$8 : 5 + 2 + 1$	$125, 50, 25$	$52, 12$
$8 : 5 + 1 + 1 + 1$	$225, 45, 45, 45$	115
$10 : 5 + 4 + 1$	$105, 84, 21$	$54, 145$
$10 : 5 + 3 + 2$	$105, 63, 42$	53

For the secondary two-block checks it is convenient to normalize by a common power of 10. For positive decimal blocks A, B and an integer relative shift d, put

$$F_d(A,B) = \begin{cases} A + 10^d B, & d \geq 0, \\ 10^{-d}A + B, & d < 0. \end{cases}$$

After the normalization in the definition of F_d, if the two shifted blocks are disjoint, lemma 2.2 gives $s(F_d) = s(A) + s(B)$ automatically. Hence only the finitely many shifts for which the two product blocks overlap need inspection. The distinct-position condition on the original digits removes the shifts in which an added digit would occupy a position already used by the displayed motif. The table below lists *every admissible overlapping shift*; entries on the same side of a semicolon have the indicated common digit sum.

Branch	(A, B)	target	admissible overlapping shifts and digit sums	exceptional input
$8 : 16 + 1$	$(2000, 125)$	10	$d = -2, -1, 2, 3 :$ 10	none
$8 : 52 + 1$	$(110500, 2125)$	17	$d = -3, -2, -1, 3, 4, 5 :$ 17; $d = 2 :$ 8	152
$8 : 12 + 5$	$(27000, 11250)$	18	$d = -4, -3, -2, -1, 3, 4 :$ 18; $d = 2 :$ 9	512
$8 : 115 + 1$	$(43125, 375)$	30	$d = -2, 4 :$ 30; $d = -1, 3 :$ 21	1151, 1115
$10 : 54 + 1$	$(1296, 24)$	24	$d = -1, 2, 3 :$ 24	none
$10 : 53 + 2$	$(1272, 48)$	24	$d = -1, 3 :$ 24; $d = 2 :$ 15	253

For example,

$$F_2(110500, 2125) = 323000, \qquad F_2(27000, 11250) = 1152000, \qquad F_2(1272, 48) = 6072,$$

with digit sums 8, 9, 15 respectively; these are exactly the three positive-shift secondary exceptions in the table. The two exceptional 115 placements give

$$F_{-1}(43125, 375) = 431625, \qquad F_3(43125, 375) = 418125,$$

both of digit sum 21. They correspond to the input words 1151 and 1115.

The terminal leaves are then direct arithmetic:

$$\begin{aligned} 125 \cdot 152 &= 19000, & 125 \cdot 512 &= 64000, \\ 1750 \cdot 1151 &= 2014250, & 2750 \cdot 1115 &= 3066250, \\ 12 \cdot 145 &= 1740, & 12 \cdot 253 &= 3036, \end{aligned}$$

whose digit sums are respectively 10, 10, 14, 22, 12, 12. Every target is larger than the original 8 or 10. The primary-motif table, the complete overlapping-shift table, and block separation outside those finite ranges therefore establish the claimed local completeness without any search over the ambient integer n.

C Local arithmetic for the pure-power eleven-family

This appendix makes explicit why the two residue classes in equations (39) and (40) are exhaustive. The residue table in the main text gives $W_r \in \{81, 63, 45, 27, 09, 90, 72, 54, 36, 18\}$ and $T_r \in \{82, 64, 46, 28, 10, 92, 74, 56, 38, 20\}$. Since $q_t = 100^{t+2}$, the lower correction at the right endpoint changes the repeated block of zero-based index $t + 1$ (the $(t + 2)$-nd block from the low end) from W_r to $W_r + 2$. The ten changes are

W_r	81	63	45	27	09	90	72	54	36	18
$s(W_r + 2) - s(W_r)$	2	2	2	2	−7	2	2	2	2	−7.

Thus the exceptional blocks are exactly 09 and 18, corresponding to $r = 5, 10$.

At the left endpoint the additional subtraction of 2 acts only on the two-digit tail. The changes are

T_r	82	64	46	28	10	92	74	56	38	20
$\mathrm{s}(T_r - 2)$	8	8	8	8	8	9	9	9	9	9.

Combining these rows with h_r and r gives exactly two possible correction constants:

	$h_r + \mathrm{s}(T_r) + \mathrm{s}(W_r + 2) - 9 - r$	$h_r + \mathrm{s}(T_r - 2) + \mathrm{s}(W_r + 2) - 9 - r$
$r \notin \{5, 10\}$	11	9
$r \in \{5, 10\}$	−7	0.

This is the complete local arithmetic used in the proof. There is no further carry state, because $W_r + 2 < 100$, and no further borrow state, because every $T_r \geq 10$.

D Formal theorem map

This map is provided for navigation of the supplementary source and is not part of the logical dependency of the manuscript. The Carry-Obstruction motif tables for 8 and 10 are the human proof route; Lean establishes the same nonmaximality conclusions through the independent certificate-soundness route described above. The principal correspondences are:

Mathematical statement	Lean declaration / file
Separated decimal blocks	`digitSum10_append` `Basic.lean`
Digit-sum subadditivity	`digitSum10_add_le` `Basic.lean`
Decimal rescaling	`larger_good_of_witness` `CarryObstruction/Theory.lean`
Carry certificate soundness	`carryObstruction_of_valid_certificate` `CarryObstruction/Certificate.lean`
Small carry obstruction	`carryObstruction_of_forbiddenSmall` `CarryObstruction/Forbidden.lean`
Defect monotonicity	`defect_add_period_le` `PeriodicCrossing/Defect.lean`
Crossing criterion	`maxGood_of_crossings` `PeriodicCrossing/Defect.lean`
Periodic block normal form	`periodicN_mul_blockForm` `PeriodicCrossing/Blocks.lean`
Aligned periodic-block update	`digitSum10_repeatBlock_update` `PeriodicCrossing/AlignedUpdate.lean`

Infinite 7 family	`N7Family_maxGood`, `N7Family_not_decimal_scaled`,
	`infinitely_many_maxGood_seven`
	`PeriodicCrossing/SevenFamily.lean`

Infinite 11 family	`N11Family_maxGood`, `N11Family_not_decimal_scaled`,
	`infinitely_many_maxGood_eleven`
	`PeriodicCrossing/ElevenFamily.lean`

Final spectrum	`small_value_spectrum_iff` `Main.lean`